

Weekly Implementation Plan

Project: Visual Similarity-Based Recommender Using Deep Learning

Mitacs GRI: Haseeb Asif

Project Description: The goal of this project is to develop a visual similarity-based recommender system that enhances product discovery by leveraging deep learning on product images. Students will explore how visual features such as style, color, shape, and design can be used to recommend similar or complementary products in ecommerce settings. The project aims to apply convolutional neural networks and multimodal models to extract meaningful image embeddings and build an efficient similarity search framework. Another goal is to investigate the use of generative models to create synthetic product samples that improve recommendation coverage, especially for underrepresented or niche catalog items. The project also seeks to evaluate how visual recommendation tools can improve user engagement, browsing experience, and personalized shopping interactions.

Week 1 – Literature Review & Project Setup

Goals

- Understand visual recommender systems
- Study image embeddings
- Install software environment
- Select dataset

Tasks

- Read 6–8 research papers
- Install Python environment
- Install PyTorch
- Download dataset
- Explore dataset structure

Deliverables

- Literature review (8 to 10 research papers), findings documented in excel sheet
- Environment setup
- Dataset summary

Week 2 – Image Preprocessing & Feature Extraction

Goals

Prepare images for deep learning.

Tasks

- Resize images
- Normalize images
- Remove corrupted images
- Build PyTorch Dataset class
- Visualize samples

Learn

- Image augmentation
- Data loaders

Deliverables

- Clean dataset
- Data preprocessing pipeline

Week 3 – CNN Feature Extraction

Goals

Extract visual embeddings.

Tasks

Implement

- ResNet50
- EfficientNet

Extract

- 512-D or 2048-D embeddings

Visualize embeddings using PCA or t-SNE.

Deliverables

- Image embedding database
- Embedding visualization

Week 4 – Visual Similarity Search

Goals

Recommend visually similar products.

Tasks

- Compute cosine similarity
- Build FAISS index
- Retrieve Top-K similar products

Evaluate

- Response time
- Recommendation quality

Deliverables

Working image similarity engine

Week 5 – Multimodal Representation Learning

Goals

Improve recommendations using CLIP.

Tasks

Learn

- CLIP image encoder

- Text encoder

Generate

- Image embeddings
- Text embeddings

Compare

- CNN
- CLIP

Deliverables

Performance comparison report

Week 6 – Recommendation Engine

Goals

Develop recommendation pipeline.

Tasks

Implement

- Query-by-image
- Similar products
- Complementary products (optional)

Design recommendation API.

Deliverables

Prototype recommender

Week 7 – User Interface & Visualization

Goals

Create interactive demo.

Tasks

Develop Streamlit dashboard

Features

- Upload image
- Display similar products
- Similarity scores
- Product gallery

Deliverables

Interactive prototype

Week 8 – Synthetic Product Generation

Goals

Improve recommendation coverage.

Tasks

Study

- Stable Diffusion
- Diffusers Library

Generate

- Synthetic fashion products
- New color/style variations

Compare

Original vs synthetic catalog

Deliverables

Synthetic product dataset

Week 9 – Recommendation Improvement

Goals

Use synthetic images.

Tasks

Add generated products

Evaluate

- Catalog coverage
- Diversity
- Novelty

Experiment with

- Data augmentation
- Fine tuning

Deliverables

Improved recommender

Week 10 – Evaluation

Goals

Evaluate system.

Metrics

Recommendation

- Precision@K
- Recall@K
- mAP
- NDCG

Similarity

- Cosine similarity

Efficiency

- Search latency
- Memory usage

Visualization

- Retrieval examples
- Confusion analysis

Deliverables

Evaluation report

Week 11 – Optimization & Final Improvements

Goals

Improve performance.

Tasks

Optimize

- FAISS indexing
- Batch inference
- GPU acceleration

Optional

- Quantization
- ONNX export

Deliverables

Optimized recommender

Week 12 – Documentation & Final Presentation

Goals

Complete project.

Tasks

Prepare

- Final report
- GitHub repository
- User guide
- Demonstration video
- Presentation slides

Deliverables

- Final prototype
- Source code
- Presentation +Technical report /Conference Paper